

How do AI Data centers use and pollute water?

Posted by u/Evamme7 • 0 points • 10 comments

I understand that they do use it for cooling and pollute lots of water but I don't Understand how it gets polluted and why the water can't just be put back into the system.

Comments

u/MrWiggles • 1 points • 2026-01-30 06:53 UTC

AI centers, use evaporative cooling. The scale of AI centers, remove a none trivial amount of potable water from communities uses. The evaporated water goes back into the water cycle. It'll get deposited in few months, somewhere else.

Thats how it removes water. Which sometime gets describe as destroying the water. Its not destroyed.

Then there water exchange which due to other elements used in the cooling process, makes the spent cooling water from data centers, brackish water. This can also have high level of sentiment. This turns the local water, from potable to non potable water. It needs to be recycle because it can be safely drank again. Brackish water has limited other uses. Such as cleaning, watering, and water displays.

Though data centers does not separate out their brackish water, so it pollutes the water table. 2

u/tulleoftheman • 1 points • 2026-01-29 18:58 UTC

The water gets hot and if you put hot water back into the ecosystem it kills things.

u/Evamme7 • 1 points • 2026-01-29 19:03 UTC

So why don't they just let it cool first?

u/JustSomeGuy_56 • 1 points • 2026-01-30 01:23 UTC

in 1980 I worked at a place that used the cooling water to heat the building. we got an award from the dept of energy. but the added cost of the plumbing far exceeded any fuel savings.

u/Indemnity4 • 1 points • 2026-01-30 01:07 UTC

They do cool it, which means some evaporates.

The classic image of a nuclear power plant and the big cone/funnel thing? That is the cooling tower. It only exists to cool water.

Hot water at something like 70°C is pumped to the top, then it flows down the walls of

the cooling tower and some evaporates. That cools the rest of the water.

You lose about 1% of the volume of water for every 10°C you need to cool it.

u/tulleoftheman • 1 points • 2026-01-29 19:15 UTC

Its a LOT of water. Each data center uses 1 to 5 million gallons of water a day. You could build big tanks for it, but it would take over a week for the water in a big tank to cool to ambient temperature. So now you need 8-10 tanks, each the size of a football stadium. That takes up a ton of land and resources, and depletes water that is in natural reservoirs, and still heats the surrounding area as the heat has to go somewhere. And you can only cool to the air temperature, while natural water sources are cooler than the air because it is coming up from underground.

u/[deleted] • 1 points • 2026-01-29 18:38 UTC

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u/binomine • 4 points • 2026-01-29 18:17 UTC

Data centers don't "pollute" the water, it definitely goes right back into the system. However, that water was fresh, drinkable water and after they are done with it, it needs to be treated to return it to drinkable water.

The catch is that the current systems often don't have enough capacity to really handle how much water a data center uses, and sometimes it overwhelms both a community's ability to generate fresh water and their sewer system to pump it away and retreat it.

u/MrWiggles • 1 points • 2026-01-30 06:49 UTC

If I cant fuckin drink it, then its polluted.

u/Superior_Mirage • 1 points • 2026-01-30 06:44 UTC

This is the only real concern with water and data centers -- greedy municipalities authorizing construction their utilities can't support.

On a national scale, AI only consumes something like 1-2% of the amount that turf (e.g. lawns, golf courses, parks, etc.) does, so if water is one's concern, that's where to hit.

Which is not to say that AI doesn't have a hundred other problems that need addressing. But if we focus on nonissues like water, we'll fail to address the real problems.